

Maharashtra Public Service Commission Assessment

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Assistant Professor Computer Engineering MCA Group A

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Assistant Professor Computer Engineering MCA Group A

Section Id :	1505617
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Sub-Section Id :	15056112
Question Shuffling Allowed :	Yes
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Question Number : 1 Question Id : 150561603 Question Type : MCQ Option Shuffling : No

The probability that a number selected at random between 100 and 999 (both inclusive) will not contain the digit 7 is _____ .

Options :

1. 16/25
2. $(9/10)^3$
3. 27/75
4. 18/25

Question Number : 2 Question Id : 150561604 Question Type : MCQ Option Shuffling : No

Skewness of normal distribution is _____ .

Options :

1. Negative
2. Positive
3. 0
4. Undefined

Question Number : 3 Question Id : 150561605 Question Type : MCQ Option Shuffling : No

If $E(x) = 2$ and $E(z) = 4$, then $E(z - x) = ?$

Options :

1. 2
2. 6
3. 0
4. Insufficient data

Question Number : 4 Question Id : 150561606 Question Type : MCQ Option Shuffling : No

Which of the following mentioned standard probability density functions is applicable to discrete random variables ?

Options :

1. Gaussian Distribution
2. Poisson Distribution
3. Rayleigh Distribution
4. Exponential Distribution

Question Number : 5 Question Id : 150561607 Question Type : MCQ Option Shuffling : No

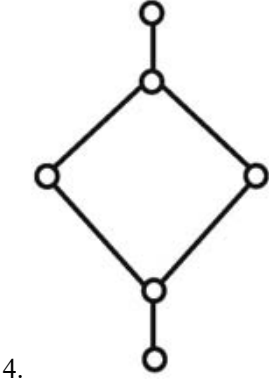
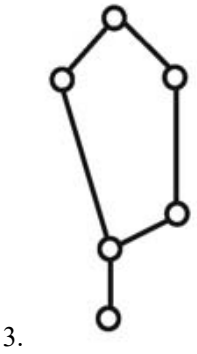
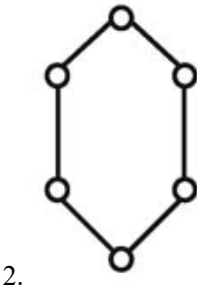
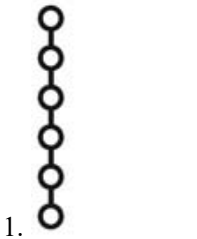
There are 15 people in a committee. How many ways are there to group these 15 people into 3, 5 and 4 ?

Options :

1. 846
2. 2468
3. 658
4. 1317

Which of the following is not a “lattice” ?

Options :



Hasse diagram is also called _____ .

Options :

1. Ordered diagram

2. Unordered diagram

3. Partial ordered diagram

4. Partial un-ordered diagram

Question Number : 8 Question Id : 150561610 Question Type : MCQ Option Shuffling : No

How many relations exist from Set X to Set Y if the Set X and Set Y has 7 and 8 elements ?

Options :

1. 256
2. 272
3. 356
4. 56

Question Number : 9 Question Id : 150561611 Question Type : MCQ Option Shuffling : No

Which of the following function is also referred to as an injective function ?

Options :

1. Many-to-one
2. Onto
3. One-to-one
4. None of the mentioned

Question Number : 10 Question Id : 150561612 Question Type : MCQ Option Shuffling : No

What is the Cartesian of Set A and Set B, if the Set $A = \{1, 2\}$ and $B = \{a, b\}$?

Options :

1. $\{(1, a), (1, b), (2, a), (b, b)\}$
2. $\{(1, 1), (2, 2), (a, a), (b, b)\}$
3. $\{(1, a), (2, a), (1, b), (2, b)\}$
4. $\{(1, 1), (a, a), (2, a), (1, b)\}$

Question Number : 11 Question Id : 150561613 Question Type : MCQ Option Shuffling : No

Which of the following binary numbers is the same as its 2's complement ?

Options :

1. 1010
2. 0101
3. 1000
4. 1001

Question Number : 12 Question Id : 150561614 Question Type : MCQ Option Shuffling : No

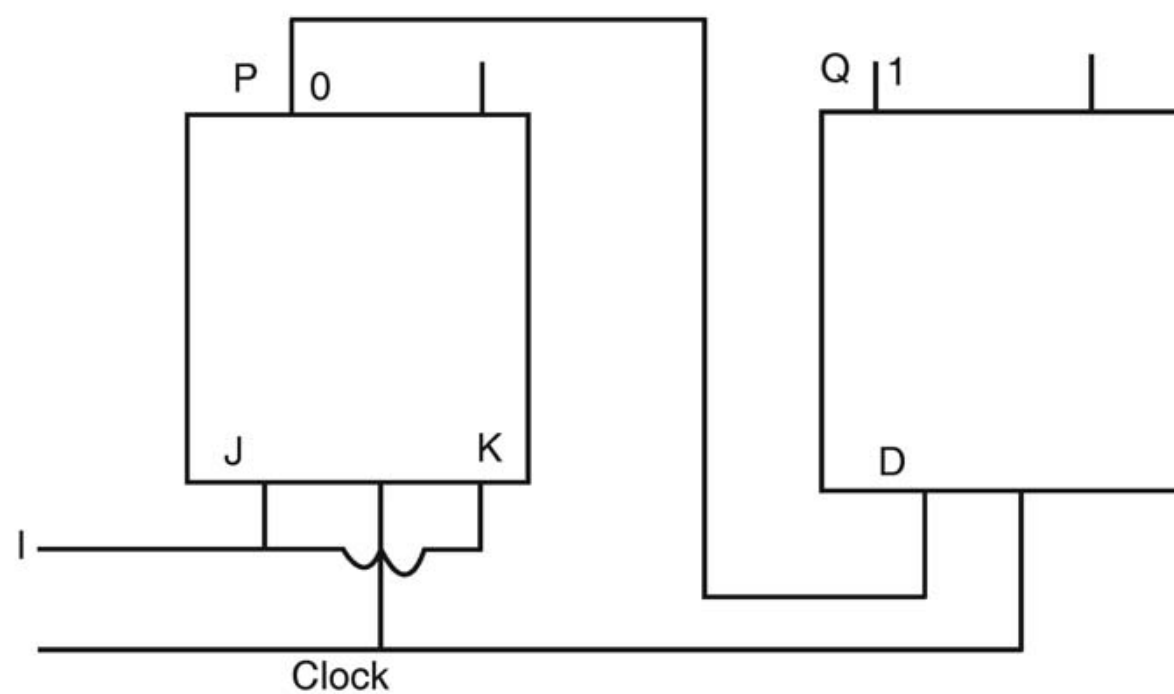
A JK flip-flop has its J input connected to logic level 1 and its input to the MQ output. A clock pulse is fed to its clock input. The flip-flop now _____ .

Options :

1. Change its state at each clock pulse
2. Go to state 1 and stay there
3. Go to state 0 and stay there
4. Retain its previous state

Question Number : 13 Question Id : 150561615 Question Type : MCQ Option Shuffling : No

The following arrangement of master slave flip-flops has the initial state of P, Q as 0, 1 respectively. After three clock cycles, the output state P, Q respectively are _____ .



Options :

1. 1, 0
2. 1, 1
3. 0, 0
4. 0, 1

Question Number : 14 Question Id : 150561616 Question Type : MCQ Option Shuffling : No

Which of the following binary coding scheme is a self-complement coding scheme ?

Options :

1. BCD
2. Excess-3
3. Biquinary
4. EBCDIC

Negative integers are generally stored in _____.

Options :

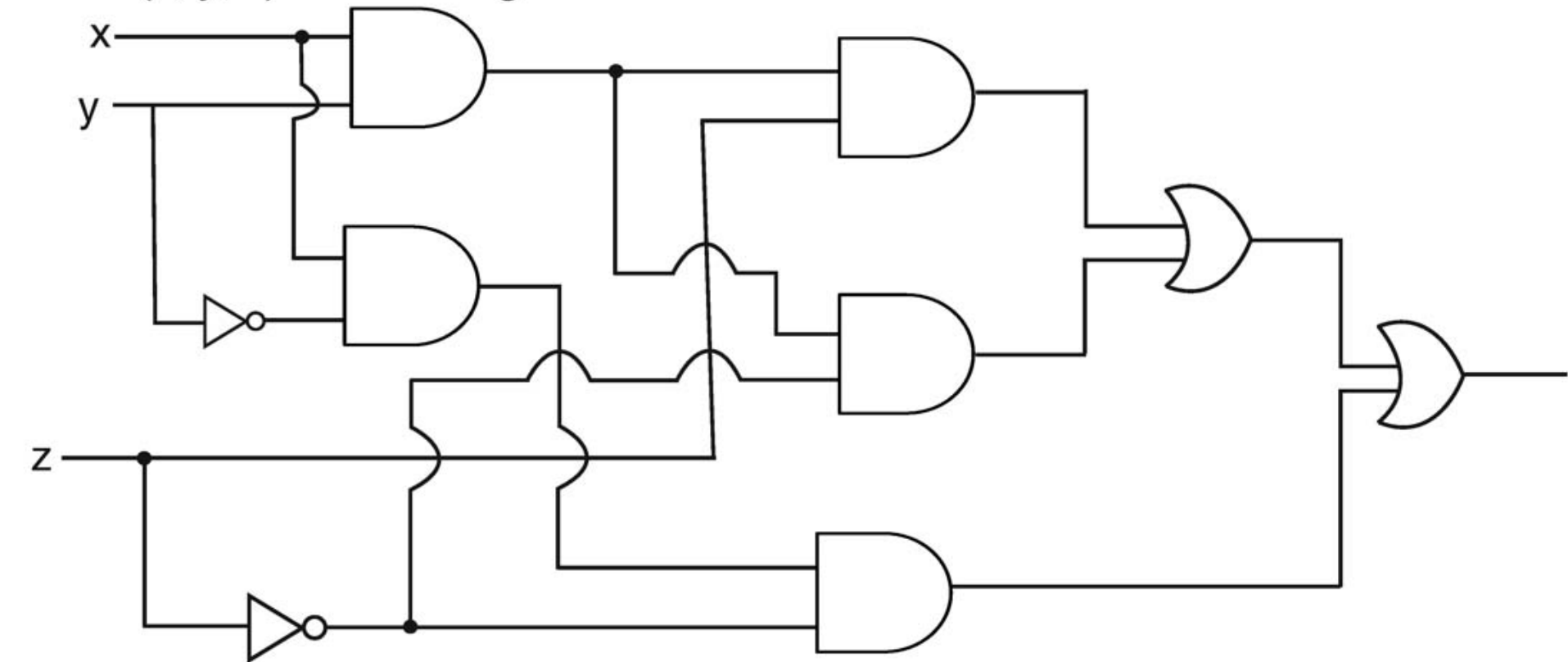
- 1's complement form
- 2's complement form
- Signed-magnitude form
- Mantissa-exponent form

Which of the following logic gates produces “0” as output when both inputs are same ?

Options :

- XOR
- NOR
- NAND
- XNOR

Find $f(x, y, z)$ for following circuit.



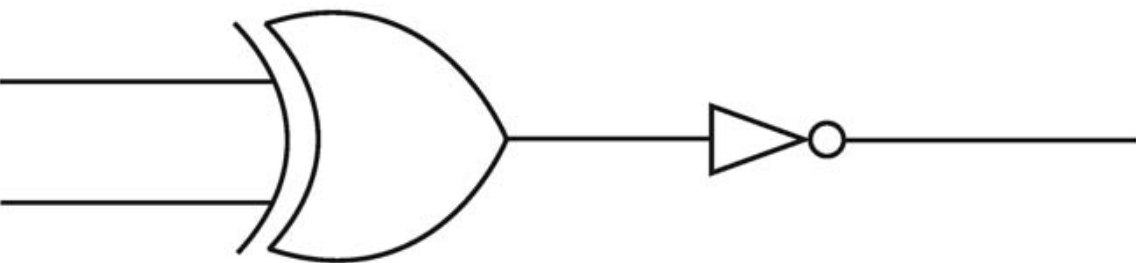
Options :

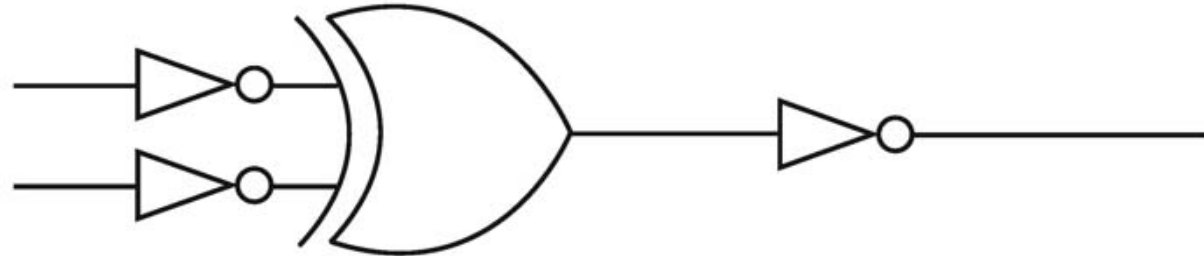
- $xyz \vee xy\bar{z} \vee x\bar{y}z$
- $xyz \vee xy\bar{z} \vee x\bar{y}\bar{z}$
- $xyz \vee xyz \vee x\bar{y}\bar{z}$
- None of the above

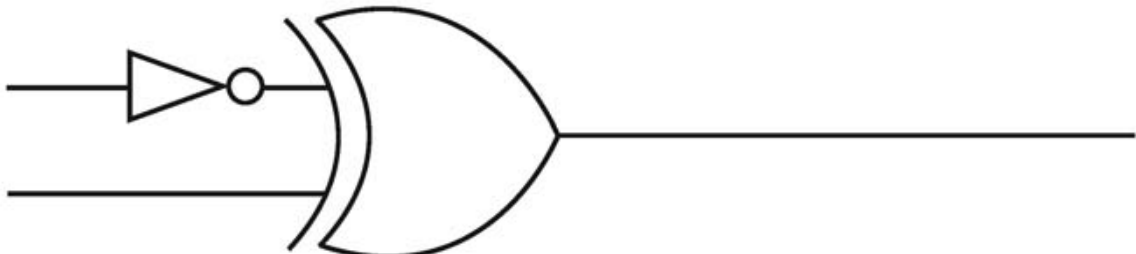
Which one of the following circuits is not equivalent to a 2-input XNOR (exclusive NOR) gate ?

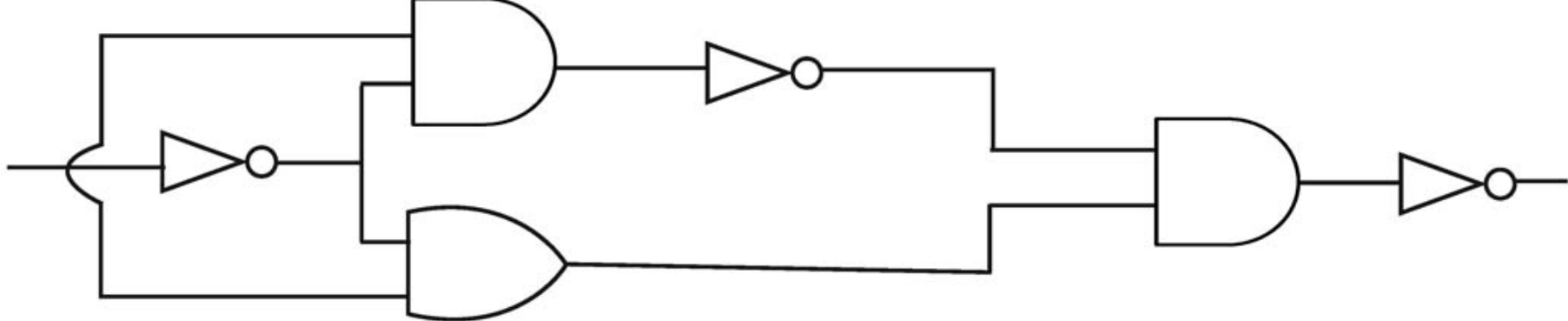
Options :

1.


2.


3.


4.



How many output lines are required for a combinational circuit to multiply two three-bit binary numbers ?

Options :

1.

2
2.

4
3.

6
4.

8

Find a minimal expansion equivalent to $xyz + x\bar{y}z + \bar{x}yz + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$ using Quine-McCluskey method.

Options :

1. $z + xy$
2. $\bar{z} + xy$
3. xy
4. $z + \bar{x}\bar{y}$

A computer uses a memory unit with 256 K words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts : an indirect bit, an operation code, a register code part to specify one of 64 registers and an address part. How many bits are there in the operation code part ?

Options :

1. 3
2. 5
3. 7
4. 9

Which of the following memory uses “Write through” technique for updating data ?

Options :

1. Cache memory and main memory
2. Main memory
3. Secondary memory
4. Tertiary memory

An instruction pipeline can be implemented by means of _____ .

Options :

1. LIFO buffer
2. FIFO buffer
3. Stack
4. None of the above

The memory locations 2000, 2001 and 2010 contains the values 25, 5 and 12 respectively before program execution. Consider following assembly program :

MOVI Rs, 2

LOAD Rd, 2000 (Rs)

ADDI Rd, 10

STOREI O(Rd), 50

Determine which memory location contains the value 50 after program execution.

Options :

1. Memory location 2000
2. Memory location 2001
3. Memory location 2010
4. Memory location 2015

A computer system has following memory hierarchy

- Cache access time = 5 ns
- Main memory access time = 50 ns
- Hit ratio = 0.92

If a memory request checks the cache first, calculate Effective Access Time (EAT).

Options :

1. 8.5 ns
2. 9.0 ns
3. 9.6 ns
4. 10.2 ns

Which of the following operations is not performed by ALU ?

Options :

1. Addition
2. AND
3. Instruction decode
4. XOR

In a 5-stage instruction pipeline, which hazard cannot be eliminated completely even with forwarding ?

Options :

1. Read After Write (RAW)
2. Write After Read (WAR)
3. Write After Write (WAW)
4. Load-use Hazard

What does the term “Fetch” refer to CPU in the context of the Fetch Decode-Execute cycle ?

Options :

1. The CPU retrieves the next instruction from memory
2. The CPU decodes the instruction
3. The CPU executes the instruction
4. The CPU stores the result

_____ provides storage internal to the CPU.

Options :

1. Main memory
2. Registers
3. Cache
4. None of the above

Question Number : 30 Question Id : 150561632 Question Type : MCQ Option Shuffling : No

With both programmed and interrupt I/O, the processor is responsible for extracting data from main memory for output and storing data in main memory for input.

What is the alternative to above situation ?

Options :

1. Direct Cache Access
2. Direct Memory Access
3. Interrupt Driven I/O
4. Optical Memory

Question Number : 31 Question Id : 150561633 Question Type : MCQ Option Shuffling : No

A full binary tree of depth K have how many nodes ?

Options :

1. $2^K - 1$
2. $2^K + 1$
3. $(2K - 1)$
4. $(2K + 1)$

Question Number : 32 Question Id : 150561634 Question Type : MCQ Option Shuffling : No

What is the correct sequence of actions for a successful PUSH operation on a STACK, assuming no overflow ?

Options :

1. Check for overflow, store the value, then increment TOP
2. Increment TOP, store the value at STACK [TOP], then check for overflow
3. Check for overflow, increment TOP, then store the value at STACK [TOP]
4. Store the value at STACK [TOP], then decrement TOP and finally check for overflow

In the context of the C language, which of the following scenarios best demonstrates the concept of call-by-reference ?

Options :

1. A function that only receives the name of the original variable and uses that name to look up the value(s) in a global symbol table
2. A function that takes integer values as arguments, performs a calculation and returns a new integer value
3. A function that declares and uses a local copy of a variable passed to it ; with the original variables value remaining the same after the function returns
4. A function that uses a pointer to access and modify the actual variable of the calling function, with changes reflecting in the original variable

Which of the following insert/delete policy used in stack ?

Options :

1. FIFO
2. Random access list
3. Sorted dynamic list
4. LIFO – insert/delete at the same end

A static variable is one _____ .

Options :

1. Which is initialized at the commencement of execution and cannot be changed at run time
2. Which is same as an automatic variable but is placed at the head of the program
3. Retains its value throughout the life of the program
4. Which cannot be initialized

Which of the following is most suitable data structure for reversing the string ?

Options :

1. Stack
2. Queue
3. Linked list
4. Heap

Which representation is most efficient for BFS on sparse graph ?

Options :

1. Adjacency matrix
2. Adjacency list
3. Edge list
4. Index table

The graph G is defined as an ordered set (V, E) then, what do the symbols $V(G)$ and $E(G)$ primarily represent ?

Options :

1. $V(G)$ is the set of vertices (nodes) and $E(G)$ is the set of edges
2. $V(G)$ is the set of graph values and $E(G)$ is the set of exponents
3. $V(G)$ is the total volume of the graph and $E(G)$ is the total number of elements in the graph
4. $V(G)$ is the set of the vertical lines and $E(G)$ is the set of endpoint coordinates

What would be asymptotic time complexity to insert an element at the second position in the linked list if pointer currently at first position ?

Options :

1. $O(n)$
2. $O(n \log n)$
3. $O(1)$
4. $O(n^2)$

A graph which has two independent sets of vertices A and B, with edges defined only between vertices of A and vertices of B is called a _____ .

Options :

1. Complete graph
2. Planar graph
3. Non-planar graph
4. Bipartite graph

A hash table of size $m = 100$, stores $n = 70$ keys (load factor $\alpha = 0.7$) using linear probing. The expected number of comparisons for a successful search is _____ .

Options :

1. $\frac{1}{1 - \alpha}$
2. $1 + \frac{\alpha}{2}$
3. $\log \frac{1}{1 - \alpha}$
4. n

Question Number : 42 Question Id : 150561644 Question Type : MCQ Option Shuffling : No

Does binary search use the divide and conquer strategy ?

Options :

1. Yes
2. No
3. Partially
4. None of the above

Question Number : 43 Question Id : 150561645 Question Type : MCQ Option Shuffling : No

The master theorem is used mostly in which of the following strategy ?

Options :

1. Divide and conquer strategy
2. Dynamic programming strategy
3. Backtracking strategy
4. Greedy strategy

Question Number : 44 Question Id : 150561646 Question Type : MCQ Option Shuffling : No

If an algorithm has running time $T(n) = 8T(n/2) + n^3$, then what is the value of $T(n)$?

Options :

1. $T(n) = \Theta(n^2 \log_2 n)$
2. $T(n) = \Theta(n^2)$
3. $T(n) = \Theta(n^3)$
4. $T(n) = \Theta(n^3 \log_2 n)$

To find the second largest one of a list of n numbers, we need atleast _____ comparisons.

Options :

1. $n - 2 - \lceil \log n \rceil$
2. $n - 2 + \lceil \log n \rceil$
3. $n + 2 + \lceil \log n \rceil$
4. $n - \lceil \log n \rceil$

What is the time complexity of the greedy algorithm to generate an optimal 2-way merge tree ?

Options :

1. $O(n \log n^2)$
2. $O(\log n)$
3. $O(n \log n)$
4. $O(\log n^2)$

Dijkstra's algorithm is greedy. When edge weights are negative, which method correctly finds single source shortest paths ?

Options :

1. Use Dijkstra but process vertices in reverse order
2. Apply fractional knapsack before Dijkstra
3. Use Bellman Ford
4. Use DFS with priority queue

In a depth-first traversal of a graph G with n vertices, k edges are marked as tree edges. The number of connected components in G is _____ .

Options :

1. k
2. $k + 1$
3. $n - k - 1$
4. $n - k$

Is merge sort a stable sorting algorithm ?

Options :

1. Yes
2. No
3. Partially
4. None of the above

Let G be a weighted undirected graph and e be an edge with maximum weight in G . Suppose there is a minimum weight spanning tree in G containing the edge e . Which of the following statements is always true ?

Options :

1. There exists a cutset in G having all edges of maximum weight
2. There exists a cycle in G having all edges of maximum weight
3. Edge e cannot be contained in a cycle
4. All edges in G have the same weight

The statement, “A Turing Machine can’t solve halting problem for all cases”
is _____ .

Options :

1. True
2. False
3. Still at open question
4. All of these

If k is the number of states of the NFA, it has _____ subsets of states.

Options :

1. $2k$
2. 0^k
3. 2^k
4. k

A language is called a regular language if same _____ recognizes it.

Options :

1. Turing machine
2. Regular expression
3. Finite automation
4. None of the above

Which part of the pumping lemma for context free languages guarantees the existence of a decomposition xyz of a string in a context free language ?

Options :

1. $|xy| \leq p$
2. $|y| > 0$
3. $xy^iz \in L$ for all $i \geq 0$
4. All of the above

A language is decidable exactly when both it and its complement are turing _____ .

Options :

1. Unrecognizable
2. Recognizable
3. Decidable
4. Undecidable

Consider a grammar :

$$G = \{ \{S\}, \{0, 1\}, P, S \}$$

Where elements of P are :

$$S \rightarrow SS$$

$$S \rightarrow 0S1$$

$$S \rightarrow 1S0$$

$$S \rightarrow \text{empty}$$

The grammar will generate _____ .

Options :

1. Regular language
2. Context-free language
3. Context-sensitive language
4. Recursive enumerable language

Which of the following statements about context-free language is false ?

Options :

1. Context-free languages are closed under string reversal
2. Context-free languages are closed under concatenation
3. Context-free languages are closed under homomorphism
4. Context-free languages are closed under set intersection

How many strings of length less than 4 contains the language described by the regular expression $(c + d)^*d(a + ab)^*$?

Options :

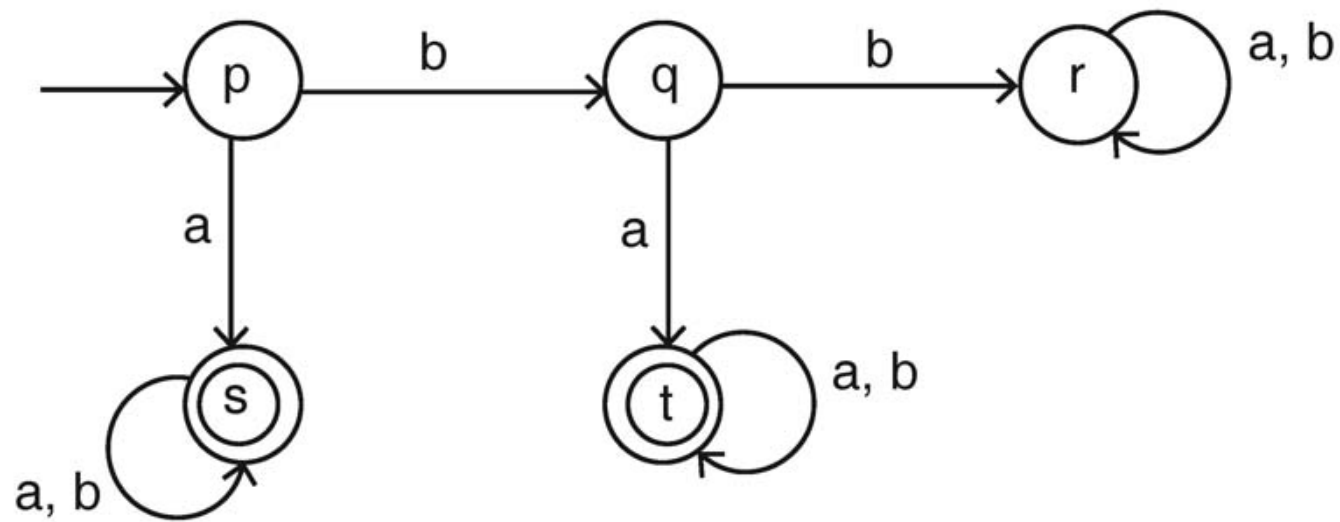
1. 11
2. 12
3. 10
4. 7

Which of the following languages are recognized by both a Finite Automation (FA) and a Turing Machine (TM) ?

Options :

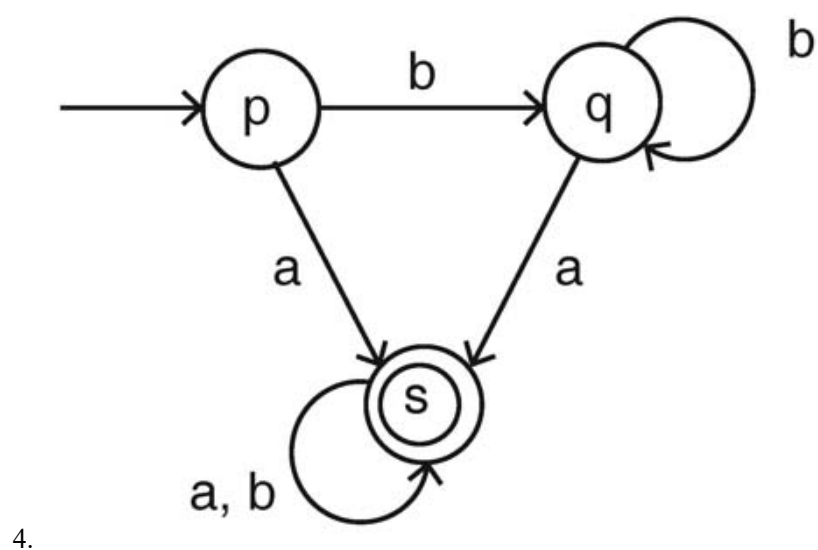
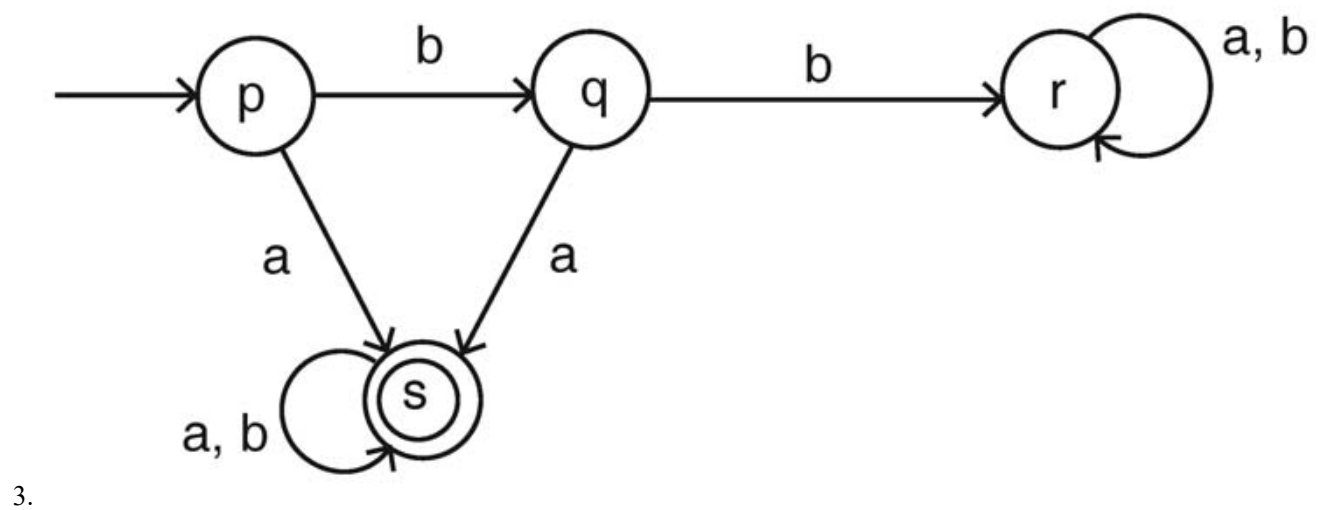
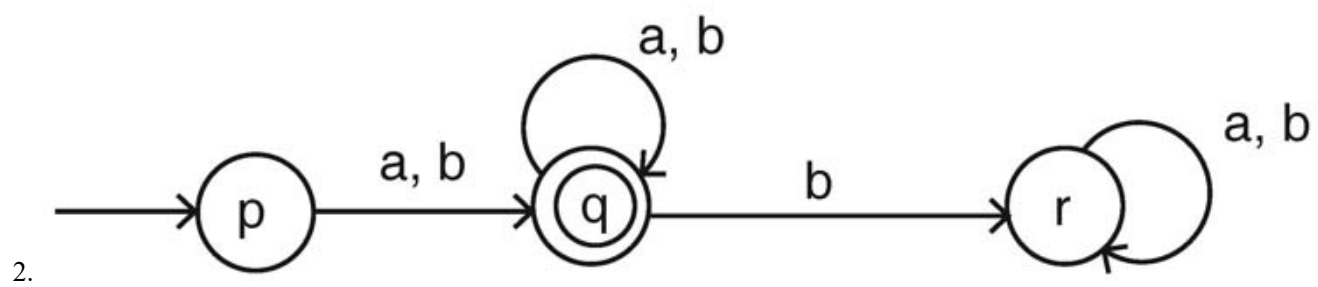
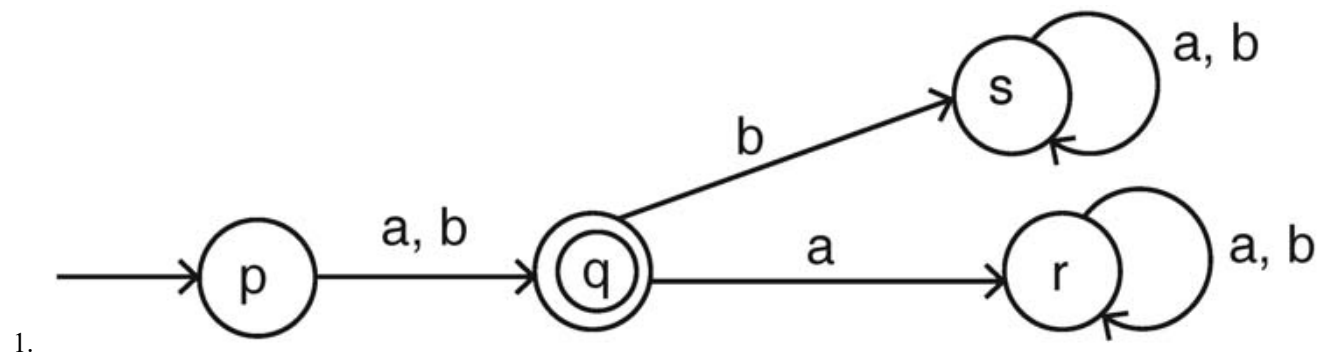
1. The set of all regular languages
2. The set of all context free languages
3. The set of all recursively enumerable languages only
4. The set of all non-regular languages only

A Deterministic Finite Automation (DFA) D with a alphabet $\{a, b\}$ is given below.



Which of the following Finite state machine is a valid minimal DFA which accept the same language as D ?

Options :



Which parsing method builds the parse tree from leaves towards the root ?

Options :

1. Top-down parsing
2. LL(1) parsing
3. Bottom-up parsing
4. Recursive-descent parsing

What is primary role of the lexical analyzer in a compiler ?

Options :

1. Construct parse trees
2. Group characters into lexemes and produce tokens
3. Generate three-address code
4. Perform semantic analysis

What is purpose of the run-time stack in a compiler's execution environment ?

Options :

1. To store machine code
2. To allocate space for local variables and activation records
3. To optimize intermediate code
4. To handle register allocation

Question Number : 64 Question Id : 150561666 Question Type : MCQ Option Shuffling : No

A set of relocatable object modules can be linked together and loaded for evaluation by _____ .

Options :

1. Compiler
2. Interpreter
3. Linking loader
4. None of the above

Question Number : 65 Question Id : 150561667 Question Type : MCQ Option Shuffling : No

A programmer writes a program to add two numbers instead of subtracting them by mistake, how can this error be detected ?

Options :

1. By interpreter only
2. By compiler only
3. By both compiler and interpreter
4. None of the above

Question Number : 66 Question Id : 150561668 Question Type : MCQ Option Shuffling : No

What is three-address code classified as ?

Options :

1. Machine code
2. High-level source
3. Intermediate representation
4. Token sequence

Consider the intermediate code given below.

- | | |
|----------------------------|-------------------|
| i. $i = 1$ | ii. $j = 1$ |
| iii. $t1 = 5*i$ | iv. $t2 = t1 + j$ |
| v. $t3 = 4*t2$ | vi. $t4 = t3$ |
| vii. $a[t4] = -1$ | viii. $j = j + 1$ |
| ix. if $j \leq 5$ goto (3) | x. $i = i + 1$ |
| xi. if $i < 5$ goto (2) | |

The number of nodes and edges in the control-flow-graph constructed for the above code, respectively, are _____ .

Options :

1. 5 and 7
2. 6 and 7
3. 5 and 5
4. 7 and 8

A grammar will be meaningless _____ .

Options :

1. If terminal set and non-terminal set are not disjoint
2. If left hand side of a production is a single terminal
3. If left hand side of a production has non-terminal
4. All of these

Which of the following statement is true ?

Options :

1. SLR parser is more powerful than LALR
2. LALR parser is more powerful than canonical LR parser
3. Canonical LR parser is more powerful than LALR parser
4. The parsers SLR, canonical LR and LALR have the same power

Type checking is normally done during _____ phase.

Options :

1. Code optimization
2. Lexical analysis
3. Syntax directed translation
4. Syntax analysis

A page fault occurs _____ .

Options :

1. When the page is not in the memory
2. When the page is in the memory
3. When the process enters the blocked state
4. When the process is in the ready state

Which of the following outcomes is/are possible when threads T_1 and T_2 execute concurrently ?

- a. T_2 runs first and prints 1, T_1 does not print anything (deadlock)
- b. T_1 runs first and prints 1, T_2 runs next and prints 2
- c. T_2 runs first and prints 1, T_1 runs next and prints 2
- d. T_1 runs first and prints 1, T_2 does not print anything (deadlock)

Options :

1. a
2. b
3. a and b
4. c and d

An operating system implements a policy that requires a process to release all resources before making a request for another resource. Select the TRUE statement from the following _____ .

Options :

1. Both starvation and deadlock can occur
2. Starvation can occur but deadlock cannot occur
3. Starvation cannot occur but deadlock can occur
4. Neither starvation nor deadlock can occur

Question Number : 74 Question Id : 150561676 Question Type : MCQ Option Shuffling : No

Which of the following tasks is/are the responsibility/responsibilities of the Memory Management Unit (MMU) in a system with paging-based memory management ?

- a. Allocate a new page table for a newly created process.
- b. Raise a trap when a virtual address is not found in the page table.
- c. Raise a trap when a process tries to write to a page marked with read-only permission in the page table.
- d. Translate a virtual address to a physical address using the page table.

Options :

1. a, b, c
2. b, c, d
3. a, c, d
4. a, b, d

Question Number : 75 Question Id : 150561677 Question Type : MCQ Option Shuffling : No

In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frame increases ?

Options :

1. LRU (Least Recently Used)
2. OPT (Optimal Page Replacement)
3. MRU (Most Recently Used)
4. FIFO (First In First Out)

In a file allocation system, which of the following allocation scheme (s) can be used if no external fragmentation is allowed ?

- a. Contiguous
- b. Linked
- c. Indexed

Options :

- 1. a and c only
- 2. b only
- 3. c only
- 4. b and c only

Consider the following table of arrival time for three processes P0, P1, P2.

Process	Arrival time	Burst time
P0	0 ms	9 ms
P1	1 ms	4 ms
P2	2 ms	9 ms

The preemptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes ?

Options :

- 1. 5.0 ms
- 2. 4.33 ms
- 3. 6.33 ms
- 4. 7.33 ms

Question Number : 78 Question Id : 150561680 Question Type : MCQ Option Shuffling : No

A counting semaphore was initialized to 7. Then 5 P (wait) operations and 3 V (signal) operations were completed on this semaphore. The resulting value of semaphore is _____ .

Options :

1. 3
2. 6
3. 5
4. 9

Question Number : 79 Question Id : 150561681 Question Type : MCQ Option Shuffling : No

Consider a hierarchical file system. If a user attempts to delete a non empty directory using 'delete directory' command without any recursive options, what is the most likely outcome ?

Options :

1. The operating system will automatically delete all files and subdirectories within it
2. Only directory structure will be removed leaving files and subdirectories orphaned
3. The operating system will prompt the user for confirmation to delete each item within the directory
4. The operating system will return an error indicating that directory is not empty

Question Number : 80 Question Id : 150561682 Question Type : MCQ Option Shuffling : No

Paging _____ .

Options :

1. Solves the memory fragmentation problem
2. Allows modular programming
3. Allows structured programming
4. Avoids deadlock

In RDBMS, different classes of relations are created using _____ technique to prevent modification anomalies.

Options :

1. Functional dependencies
2. Data integrity
3. Referential integrity
4. Normal forms

Consider the following table :

Faculty (facName, dept, office, rank, dateHired)

facName	dept	office	rank	dateHired
Ravi	Art	A 101	Professor	1975
Murali	Math	M 201	Assistant	2000
Narayanan	Art	A 101	Associate	1992
Lakshmi	Math	M 201	Professor	1982
Mohan	CSC	C 101	Professor	1980
Lakshmi	Math	M 201	Professor	1982
Sreeni	Math	M 203	Associate	1990
Tanuja	CSC	C 101	Instructor	2001
Ganesh	CSC	C 105	Associate	1995

(Assume that no faculty member within a single department has same name. Each faculty member has only one office identified in office.)

Then Faculty is _____ .

Options :

1. Not in 3NF, not in BCNF
2. In 3NF, in BCNF
3. Not in 3NF, but in BCNF
4. In 3NF, not in BCNF

Under Two-Phase Locking (2PL) :

- Growing Phase = acquiring locks
- Shrinking Phase = releasing locks

When can releasing begin ?

Options :

1. After first lock
2. After all required locks acquired
3. After commit only
4. After releasing shared locks

A relation $R = \{A, B, C, D, E, F, G\}$ is given with following set of functional dependencies :

$F = \{AD \rightarrow E, BE \rightarrow F, B \rightarrow C, AF \rightarrow G\}$

Which of the following is a candidate key ?

Options :

1. A
2. AB
3. ABC
4. ABD

Let E_1 and E_2 be two entities in an E/R diagram with simple single valued attributes. R_1 and R_2 are two relationships between E_1 and E_2 , where R_1 is one-to-many and R_2 is many-to-many. R_1 and R_2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model ?

Options :

1. 2
2. 3
3. 4
4. 5

Given :

```
CREATE TABLE department (  
dept-name VARCHAR (20) PRIMARY KEY) ;  
CREATE TABLE Course (  
Course_id VARCHAR (8), dept_name VARCHAR (20),  
FOREIGN KEY (dept_name) REFERENCES  
department );  
What happens if a non-existent dept-name is inserted ?
```

Options :

1. Department auto-created
2. Insert succeeds with warning
3. Insert fails due to referential integrity
4. dept_name becomes NULL

Which integrity constraints allows attributes to form a candidate key but permits them to be null unless explicitly declared otherwise ?

Options :

1. The not null constraint
2. The primary key constraint
3. The unique constraint
4. The check constraint

The join operation can be defined as _____ .

Options :

1. A cartesian product of two relations followed by selection
2. A cartesian product of two relations
3. A union of two relations followed by cartesian product of two relations
4. A union of two relations

What is the term used to describe the participation of an entity set E in a relationship set R when only some entities in E participate in relationship in R ?

Options :

1. Mandatory participation
2. Partial participation
3. One-to-Many cardinality
4. Total participation

A relation R(A, B, C) has dependencies $A \rightarrow B$ and $B \rightarrow C$. Which situation makes it NOT in 3NF ?

Options :

1. A is a super key
2. C is a prime attribute
3. B is not a super key and C is non-prime
4. All attributes are part of some key

What is the function of twisting in twisted pair cable ?

Options :

1. To make it probable that both wires are equally affected by noise or crosstalk
2. To increase speed of data transfer
3. To increase data transfer capacity
4. To transfer the data at longer distances

A page of text document has average 24 lines of text, each line can have 80 characters. We need to download text document at rate of 1000 pages per second. What should be required bit rate of the channel if one character requires 8 bit ?

Options :

1. 15.36 Mbps
2. 1.92 Mbps
3. 11.36 Mbps
4. 8.36 Mbps

Which of the following data units has an application layer message plus header from 4th layer in TCP/IP protocol suite ?

Options :

1. Frame
2. Datagram
3. Segment/User datagram
4. Packet

If the frame buffer has 8 bits per pixel and 8 bits allocated for each of the R, G, B components, what would be size of the lookup table ?

Options :

1. 24 bytes
2. 1024 bytes
3. 768 bytes
4. 256 bytes

An ISP has following chunk of CIDR-based IP addresses available with it : 245.248.128.0/20. The ISP wants to give half of this chunk of addresses of organization A and a quarter to organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B ?

Options :

1. 245.248.136.0/21 and 245.248.128.0/22
2. 245.248.128.0/21 and 245.248.128.0/22
3. 245.248.132.0/22 and 245.248.132.0/21
4. 245.248.136.0/24 and 245.248.132.0/21

The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted as _____ .

Options :

1. 11001001000
2. 11001001011
3. 11001010
4. 110010010011

Consider a parity check code with three data bit and four parity check bit. Three of the code words are 0101011,10001101 and 1110001, which of the following are also code words ?

- I. 0010111
- II. 0110110
- III. 1011010
- IV. 0111010

Options :

- 1. I and III
- 2. I, II, III
- 3. II and IV
- 4. I, II, III and IV

Suppose computers A and B have IP address 10.105.1.113 and 10.105.1.91 respectively and they both use the same net mask N. Which of the values of N given below should not be used if A and B should belong to the same network ?

Options :

- 1. 255.255.255.0
- 2. 255.255.255.128
- 3. 255.255.255.192
- 4. 255.255.255.224

For stop-and-wait flow control, for n data packets sent, how many acknowledgments are needed ?

Options :

- 1. n
- 2. 2n
- 3. n – 1
- 4. n + 1

Which of the following is not a valid layer of the TCP/IP model ?

Options :

1. Transport Layer
2. Internet Layer
3. Application Layer
4. Presentation Layer